



Figure 4: Get involved in the community

interviews, etc. Often, organisations are explicit about what they expect of you. Otherwise, it's more about bringing in new ideas and perspectives.

Next comes understanding the code base. This is a daunting process if you're new. But a little perseverance can get you through this. You can always ask for help in the community discussion forums and then go on to help others whenever you can. Always start fixing minor bugs and get them reviewed. Some organisations have 'junior-jobs'

dedicated for new contributors.

Other than programmers, many projects need designers and people in support roles such as social media managers. These are roles people might not know about unless they ask those already working on the project to describe the needs of a specific project.

Finally, most communication takes place asynchronously, at different times and in different parts of the world. You shouldn't expect immediate responses. Be patient with other contributors and don't be afraid to follow up if no one responds to your query. And in these communities, open conversation and a diverse range of viewpoints are encouraged, which improves both the open source software or tools, and the community that's contributing to it.

So, get started! 

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```
plot.SetTerminal("postscript");
plot.SetLegend("X Axis", "Y Axis");
plot.AppendExtra("set xrange [-2:2]");
plot.AppendExtra("set yrange [-2:2]");
Gnuplot2dDataset dataset;
dataset.SetTitle("2-D Data");
dataset.SetStyle(Gnuplot2dDataset::LINES);
for(t = 0; t <= 10; t+=0.1)
{
    x=sin(2*t);
    y=sin(4*t);
    dataset.Add(x, y);
}
plot.AddDataset(dataset);
ofstream plotFile("graph.plt");
plot.GenerateOutput(plotFile);
plotFile.close();
return 0;
}
```

The instruction `Gnuplot plot ("graph.eps");` creates an object of the `Gnuplot` class and names the output file as `graph.eps`. This object is then used in the instruction `plot.SetTerminal("postscript");` to decide the output type. The graphic format associated with PostScript is Encapsulated PostScript with the extension `.eps`. The instruction `Gnuplot2dDataset dataset;` creates a dataset to which the source file will add data for plotting. The instruction `'dataset.`

`SetStyle(Gnuplot2dDataset::LINES);` decides the style of drawing the graph. The other styles for drawing the graph include `LINES_POINTS`, `POINTS`, `DOTS`, etc. On execution, the ns-3 script `graph.cc` (the command for execution is `./waf; ./waf --run scratch/graph`) will produce a file called `graph.plt`, which contains all the instructions required for generating the graph. You can generate the image file containing the graph `graph.eps` by executing the command `gnuplot graph.plt`. For better visualisation, I have used the functions `sin(2t)` and `sin(4t)` for generating the X and Y coordinates of the graph. Figure 2 shows the graph generated by the plot file.

We have now discussed the different ways to represent information obtained from ns-3 in a neat and concise way. What else remains to be discussed about ns-3? As I have mentioned in the very first part of this series, ns-3 is not a single tool. It is the collection of a set of utilities and languages to achieve the goal of network simulation. Before winding up this series, we still need to discuss the use of Python in ns-3 and `pcap` based tracing. But that is not all. We also need to explore the source code of ns-3 briefly and understand the infinite possibilities while expanding the source code. 

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